How The Senate Forecast Works: A Guide for Policy and Coding Students

Oct 16, 2026 · @Rylyn

Who this is for

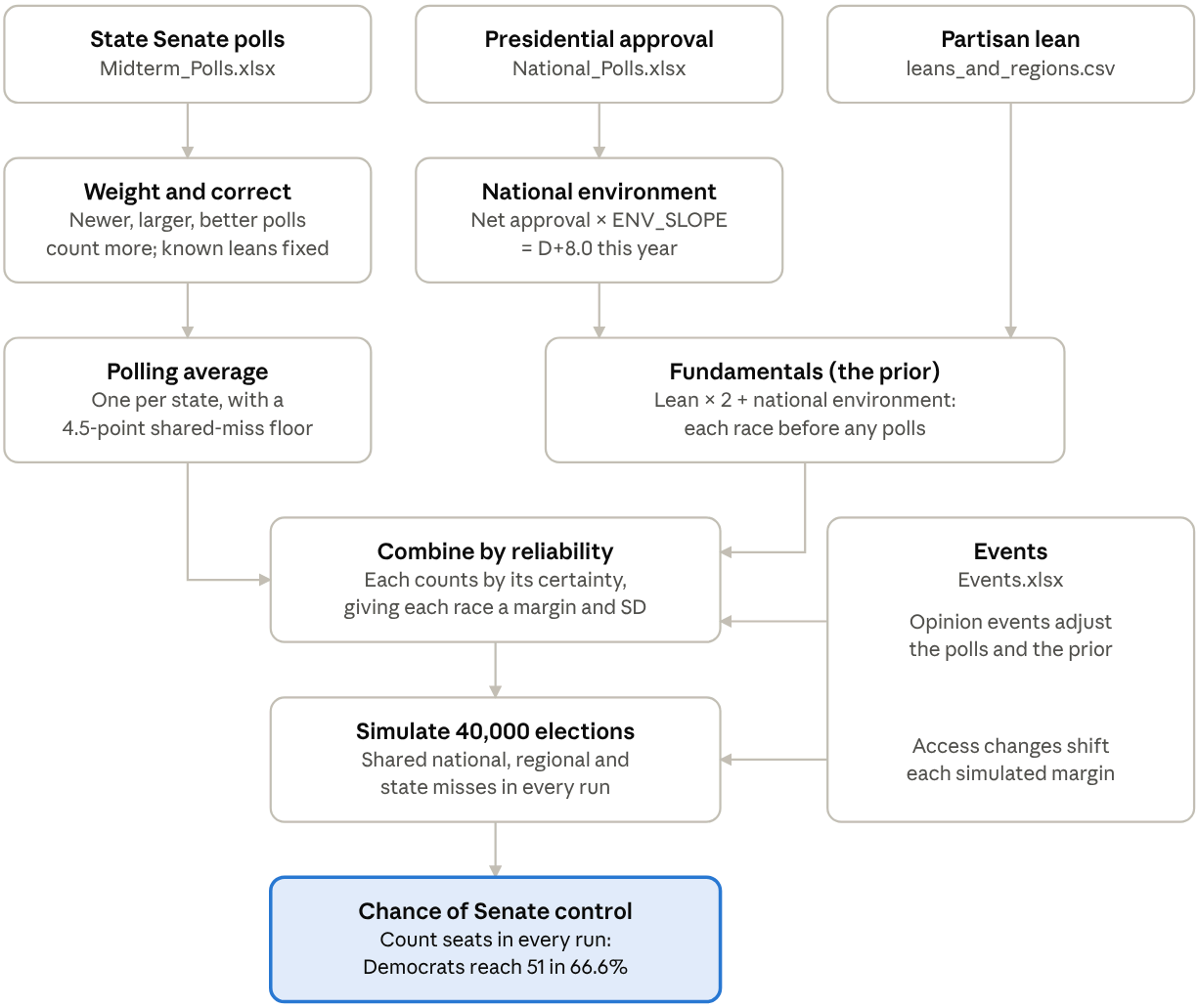
The model answers one question: what is the chance that Democrats control the Senate after November 3? Using the October 15 data, the answer is 66.6%. This guide explains how the model reaches that number. It is written for two groups who each know half of the story.

- **Policy students who don't code:** read the flow, the Iowa example and the section for policy students. You can judge most of the important choices without opening the code.
- **Coding students who don't study policy:** read the flow, the Iowa example and the section for coding students. The math is simple. The hard part is knowing which numbers political science backs up.
- **Everyone:** the vocabulary table and "Who decides what" show how the two groups split the work.

All the margins in this guide are the Democrat minus the Republican, in percentage points. D+3 means the Democrat leads by 3 points. R+3 means the Republican leads by 3, which the code stores as −3. Each setting, its tested range and how much it moves the result are listed in Senate Forecast Model: Variables and Choices.

The model at a glance

Polls and fundamentals meet in one step; simulation turns that into odds



senate_event_model.py, from data files to result · 4 inputs, 7 steps

Read the diagram from top to bottom. The left column is the polls. The middle and right columns are the fundamentals. The events come in from the side. Everything meets at one step, "Combine by reliability," which is why the settings that decide that step matter most.

One race, start to finish: Iowa

Iowa begins as 21 polls averaging D+0.5 and ends with Democrats at a 44% chance to win. It's a good race to follow because the polls and the fundamentals disagree, and the model has to decide between them.

Step	What happens	Iowa on October 15
1. Collect polls	Every Iowa Senate poll in `Midterm_Polls.xlsx`	21 polls, August 31 to October 15. Simple average D+0.5
2. Weight each poll	Newer, larger, better-rated polls count more. Sponsored polls, generic questions and repeat releases of one survey count less	Marist's October 15 poll counts the most. Co/Efficient's generic-ballot version counts about 1/50 as much. The 21 polls carry as much information as 11.7 equal polls
3. Correct known leans	Registered-voter polls move 1 point toward Democrats. Party-sponsored polls move 3 points away from the sponsor	Trafalgar (Republican-sponsored) goes from R+2 to D+0.9
4. Average the polls	A weighted average, with an uncertainty that never drops below the 4.5-point miss that all polls in a state can share	D+1.1, give or take 4.9 points
5. Build the fundamentals	Partisan lean × 2, plus the national environment (net approval of −22.9 × 0.35 = D+8.0)	Lean R+6 gives −12. Adding 8.0 gives R+4.0, give or take 7 points
6. Combine	Each source counts in proportion to how certain it is. The polls are more certain, so they get more weight	The polls get 67%, the fundamentals 33%: R+0.5
7. Allow for change	19 days are left, and opinion can drift 0.28 points per day (spread as $\sqrt{19}$)	Total uncertainty: 4.2 points
8. Simulate 40,000 elections	Each simulation draws a national polling miss, a regional miss and an Iowa-only miss	Democrats win Iowa in 44% of simulations
9. Count seats	Add each simulation's winners to the 34 Democratic seats not up this year	When Democrats win Iowa, they control the Senate 95% of the time. When they lose it, 44%

Notice step 6. The polls say D+1.1, the fundamentals say R+4.0, and the forecast lands in between, closer to the polls. That one compromise, repeated in each battleground, explains why the settings that set each side's uncertainty (`PRIOR_SD` and `SYSTEMATIC_POLL_ERROR_SD`) move the final number so much.

For policy students: what the code is doing

Every step in the code rests on one of five ideas. None of them needs programming to understand, and most of them can be argued about without opening the code.

A weight is a multiplier

A poll with weight 0.5 counts half as much as a poll with weight 1. Weights multiply, so a party-sponsored poll (0.5) of registered voters (0.85) counts 0.425. Each weight is a judgment about how much to trust a kind of evidence. That is a policy-style question, not a coding question.

An average is only as sure as its shared mistakes

The margin of error printed with a poll only counts sampling luck. Real polls also miss because of who answers and how the pollster models turnout, and those misses hit every poll in a state at once. So after 20 polls, the model still allows a 4.5-point shared miss. Adding a 21st poll can't remove it.

Two sources combine by how reliable each one is

The fundamentals (how the state usually votes, plus the national mood) and the polls are two witnesses. Each one counts in proportion to its reliability. A state with many recent polls leans on the polls. A state with none falls back on the fundamentals. This is the same logic as weighing a long-running program evaluation against a single new study.

Simulation is counting, not prediction magic

No simple formula gives the chance of every combination of 35 races. So the computer runs the election 40,000 times. In each run, every race gets a random miss of a realistic size. It counts how often Democrats reach 51 seats. 66.6% means they reached 51 in 26,658 of the 40,000 runs.

Races move together

If the polls overstate Democrats in Iowa, they probably do in Kansas too. The model builds this in: every run includes one national miss that all states share, one regional miss and one miss for states with similar politics. Without it, the model would treat a sweep as nearly impossible, when sweeps happen in wave years.

What 66.6% does and doesn't mean

It isn't a vote share, and it isn't a call that Democrats will win. It says the Republican outcome happens in about one run out of three. If Republicans keep the Senate, the model wasn't necessarily wrong, any more than a 33% chance of rain is wrong on a day it rains. A forecast is judged over many calls, by whether its 70% calls come true about 70% of the time.

What you can do without coding

Open `senate_event_model.py` in any text editor. The SETTINGS block at the top reads like a list, and each line has a comment that explains it. Change a number, save, and run the file. The results land in the `outputs` folder as CSV files, which open in Excel. The voting-access and opinion events live in `Events.xlsx`, so you can add or edit them without touching code.

For coding students: the political science behind the code

The code is a weighted average, a Bayesian update and a Monte Carlo loop. What decides whether it is right is the political science behind each constant. Here is the reasoning behind each step.

Partisan lean is the baseline

A state's lean is how much more Democratic or Republican it votes than the country as a whole. Most voters' party loyalty is stable, and ticket splitting has collapsed, so Senate races track the state's lean more closely than they used to (Jacobson, 2015). Lean is measured in vote share. One point of share is two points of margin, which is why the code multiplies it by 2.

Midterms are a referendum on the president

The president's party usually loses ground in midterms, and presidential approval is one of the strongest predictors of how much (Tufte, 1975). The president is a Republican with a net approval of −22.9, so the national environment favors Democrats. `ENV_SLOPE` converts approval into points of margin. Because it scales every race at once, it is the biggest single setting.

Who votes matters as much as what people think

Roughly half of eligible adults vote in a midterm. Pollsters screen for likely voters, and in 2026 those likely voters run about 1 to 2 points more Democratic than all registered voters. That is why registered-voter polls are shifted toward Democrats and count a little less.

Campaigns release the polls that help them

A party-sponsored poll is real data, but the sponsor chooses which polls to publish. The released ones lean toward the sponsor, so the code moves them 3 points the other way and gives them half weight. Firms also have steady leans from their methods, called house effects. That is why the pollster rating matters.

Candidates still matter in Senate races

Senate races are statewide and candidate-driven, so they can stray far from the state's lean. Maine shows this: the fundamentals say D+16, and the polls say D+2.3 for a long-serving Republican incumbent. `PRIOR_SD` sets how far a race is allowed to stray. The optional incumbency setting adds a fixed bonus.

Polling misses are shared

Polls have missed in the same direction across many states in the same year. In 2016 and 2020 they understated Republicans. In 2022 they understated Democrats in Senate races by a median of 2.1 points (Wang, 2022). The correlation settings exist because state errors are not independent draws.

Senate control has its own rules

35 seats are up: the regular class plus special elections in Ohio and Florida. Democrats hold 34 of the seats not up, counting the two independents who caucus with them. The vice president breaks 50-50 ties, and the vice president is a Republican, so Democrats

need 51. In Nebraska, independent Dan Osborn is challenging the Republican, and he won't say which party he'd caucus with. So a win for him counts for Democrats only with a set probability.

Voting access changes who gets counted

Polls measure what people prefer. Rules like mail ballot deadlines, polling place cuts and ID requirements change whose ballot is counted. So the model applies them after the polls. Their effect is the share affected × the share deterred × how partisan those voters are. That product is usually tiny. Strict ID laws, for example, show small average turnout effects in careful studies (Cantoni & Pons, 2021).

Redrawn maps don't touch Senate races

Senate races are statewide, so House district lines can't change their margins. The code enforces this, and maps only feed the separate House module.

Shared vocabulary

One table, so a term means the same thing in a policy discussion and in the code. The last column tells you where to find each term in `senate_event_model.py` or the data files.

Term	Plain meaning	Why it matters politically	Where it lives
Margin	Democrat's share minus Republican's, in points	Who wins and by how much	`margin`, `post_mean`
Partisan lean	How much more Democratic or Republican a state votes than the nation	Party loyalty is the most stable force in elections	`leans_and_regions.csv`, `PartisanLean`
National environment	The national Democratic margin implied by presidential approval	Midterms judge the president's party	`national_environment()`, `ENV_SLOPE`

Term	Plain meaning	Why it matters politically	Where it lives
Fundamentals (prior)	The forecast before any polls: lean × 2 + environment	What a race "should" look like for a generic candidate	`build_race_table()`, `prior_mean`, `PRIOR_SD`
Posterior	The forecast after combining the fundamentals and the polls	The model's best guess for each race	`post_mean`, `post_sd`
Likely or registered voters (LV, RV)	Polled people who are expected to vote, or everyone registered	Midterm turnout is low and uneven by party	`LVorRV`, `RV_WEIGHT`, `RV_MARGIN_SHIFT`
House effect	A pollster's steady lean toward one party	Sponsors and methods tilt results	`PARTISAN_HOUSE_EFFECT`
Generic ballot	"Democrat or Republican for Congress?" with no names	Measures the party mood without candidate effects	`Generic`, `GENERIC_WEIGHT`
Full field or head-to-head	A ballot with third-party names, or only the two major candidates	Third-party support usually fades by Election Day	`Bigger_Field`, `BIGGER_FIELD_WEIGHT`
Half-life	Age at which a poll counts half	Opinion moves, so old polls go stale	`RECENCY_HALF_LIFE_DAYS`
Effective number of polls	How many equal polls the weighted set is worth	Ten polls from one firm aren't ten independent views	`n_eff` in `polling_averages()`

Term	Plain meaning	Why it matters politically	Where it lives
Standard deviation (SD)	The typical size of a miss. About 2 in 3 outcomes fall within 1 SD	Sets how confident each race is	Settings ending in `_SD`
Systematic error	A miss shared by every poll in a state	Turnout models fail together	`SYSTEMATIC_POLL_ERROR_SD`
Drift	How far opinion can move before Election Day	Late news and late deciders	`DAILY_DRIFT_SD`
Fat tails	Big misses happen more often than a bell curve suggests	Upsets and wave years	`T_DF`
Correlation	How much states miss together	National waves hit many races at once	`CORR_NATIONAL`, `CORR_REGION`, `CORR_LEAN`, `draw_base()`
Monte Carlo simulation	Running the election many times with random misses and counting the outcomes	Turns race odds into chamber odds	`forecast()`, `simulate()`, `N_SIMS`
Seed	The starting point for the random numbers	Lets anyone reproduce the exact result	`SEED`
Holdover seats	Seats not on the ballot this year	Democrats start with 34, Republicans with 31	`HOLDOVER_D`, `HOLDOVER_R`
Caucus	The party a senator organizes with	Decides control, even for independents	`NEBRASKA_CAUCUS_PROB_D`, `D_SEATS_FOR_CONTROL`

Term	Plain meaning	Why it matters politically	Where it lives
Battleground	One of the 13 contested races	Where control is decided	`BATTLEGROUNDS`
Deciding seat	The race that gives one party its majority in a given simulation	Shows which battleground really matters	Battleground table, `battlegrounds.csv`
Opinion event	News that changes what voters want	Scandals, the economy, ad money	`Events.xlsx` (Opinion sheet), `USE_OPINION_EVENTS`
Access event	A rule change that affects whose ballot counts	Deadlines, polling places, ID, citizenship proof	`Events.xlsx` (Access sheet), `USE_ACCESS_EVENTS`
Sensitivity sweep	Rerunning the model with one setting changed at a time	Shows which judgment calls drive the result	`--sweep`, `run_sweep()`, `sensitivity.csv`

Who decides what

The choices that move the forecast most are policy judgments that need no code, only a number and a reason. The code work is making those judgments testable and keeping the data clean.

Decision	The question to answer	What it takes	Who leads
`ENV_SLOPE`	How strong is the mood against the president's party this year?	Compare with the likely-voter generic ballot and past midterms	Policy

Decision	The question to answer	What it takes	Who leads
`PRIOR_SD`, `INCUMBENCY_ADVANTAGE`	How much do candidates matter compared with party?	Knowledge of the races: Collins in Maine, Osborn in Nebraska, open seats	Policy
`D_SEATS_FOR_CONTROL`, `NEBRASKA_CAUCUS_PROB_D`	What will senators do after the election?	Senate rules, caucus statements, reporting	Policy
`RV_MARGIN_SHIFT`, `PARTISAN_HOUSE_EFFECT`	Which polls overstate which party?	Paired survey results, published forecaster methods	Both
Rows in `Events.xlsx`	Which events and rule changes matter, and for how many voters?	Research literature, state election data, court rulings	Policy, with code to check the polls
Correlation settings	Which states move together?	Regional and demographic knowledge, plus the code's factor model	Both
Data updates and cleaning	Are the polls entered correctly?	Fixing pollster names, merging survey versions, rerunning	Code
New features	Should the model handle undecided voters or blend in the generic ballot, as the group model does?	Writing and testing new functions	Code, with policy input on the rule
Checking the result	Is a change real or just noise?	`--sweep`, a fixed `SEED`, enough simulations	Code

A simple way to split the work: the policy side writes down each value with a one-sentence reason and a source. The coding side runs the sweep and reports how much each choice moved the forecast. Neither side has to learn the other's whole field to do this.

Common misreadings

Most confusion about the model comes from reading a probability as a vote share or a promise.

Misreading	Correct reading
"Democrats get 66.6% of something."	66.6% is the share of simulated elections in which Democrats reach 51 seats. It is not a vote share or a seat share.
"Iowa polls show D+1.1, so Democrats are favored there."	The forecast is R+0.5, because the fundamentals (R+4.0) pull the polls back. Democrats win Iowa in 44% of simulations.
"More polls always make the forecast surer."	After about a dozen polls, the miss that all of a state's polls share is the main source of uncertainty. More polls barely help.
"Voter suppression laws should change the result a lot."	The real 2026 changes in the model move margins by hundredths of a point. It would take about 4% of the national electorate losing their ballots to move control by 10 points.
"Redrawn House maps help one party win the Senate."	Senate races are statewide. Maps change House seats only.
"If Republicans win, the model was wrong."	A 1-in-3 outcome happens one time in three. A forecast is judged over many calls, not one.
"66.6% one run and 66.9% the next means something changed."	With 40,000 simulations, chance alone moves the result by about 0.25 points. Compare runs that use the same `SEED`.

Misreading	Correct reading
"The model with the most variables will win the workshop."	Which values you choose for a few settings matters more than how many variables you add. Four choices produce most of the swing.

References

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Wang, S. (2022, November 20). *PEC (and polling) performance review, 2022*. Princeton Election Consortium. https://election.princeton.edu/articles/pec-and-polling-performance-review-2022/

All numbers in this guide come from `senate_event_model.py`, run on the professor's October 15 data files.